

The Global Effects of Carbon Border Adjustment Mechanisms

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Carbon border adjustment mechanism (CBAM)

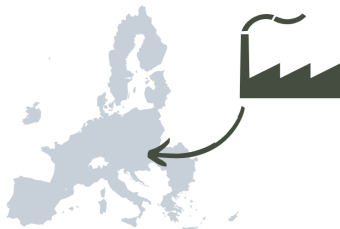


EU Production



90€

EU production is
subject to the **EU-ETS***
*(Assuming an ETS allowance
price of 90€ per tonne of CO₂)*



Non- EU Production



80€

10€

Non-EU production is
subject to a lower **ETS**
and **CBAM certificates**

Climate change is a collective action problem

- Individual countries bear the costs of carbon regulation
 - The benefits of lower emissions are shared globally
- Incomplete regulation raises two domestic concerns
 - Competitiveness: regulated producers face higher costs
 - Leakage: consumption and production shift abroad
- Can trade policy make carbon regulation easier to sustain?

This paper

- Carbon pricing creates a global green premium
 - Emissions intensity is income-neutral and cost-neutral
- CBAMs improve regulation along several margins
 - Competitiveness, leakage, abatement, and participation
- Quantitative trade model with global plant-level microdata
 - For aluminum and steel, which account for 11% of global CO₂

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CBAMs change who bears the cost of regulation

- **Competitiveness:** European producer losses fall 64%
 - From \$13.5B without a CBAM to \$4.9B with one
- **Leakage:** global emissions reductions rise 30%
 - From 37.3 Mt to 48.4 Mt of CO₂
- **Participation:** border adjustment can lower joining costs
 - By 17% to 80% for several major partners
- **Income:** lower-income countries are not systematically hit harder
 - High-emission producers bear the burden, regardless of income

Contributions

- Border adjustment can support **collective climate action**

Nordhaus (2015), Böhringer et al. (2016), Farrokhi and Lashkaripour (2025), Hsiao (2026)
Kortum and Weisbach (2017, 2026), Clausen and Wolfram (2023), Harstad (2024, 2025)

- Global equilibrium with **plant-level heterogeneity**

Böhringer et al. (2012), Larch and Wanner (2017), Mörsdorf (2022), Bellora and Fontagné (2023)
Shapiro and Walker (2018), Fowlie and Reguant (2022), Chen et al. (2025), Coster et al. (2025)

- Plant-level abatement and **policy implementation**

Fowlie (2009), Fowlie et al. (2016), Casey et al. (2026), Kortum and Weisbach (2026)
Asker et al. (2019), Domínguez-lino et al. (2026), Brunel and Levinson (2025)

Policy timeline

- **EU CBAM** proposed in 2021
 - Phase-in starting October 1, 2023 with reporting only
 - Definitive regime from January 1, 2026 for target sectors
 - Free ETS allowances phase out through 2034
- UK and Norway targeting 2027 implementation
 - EFTA carbon pricing aligned with EU and thus exempt
- Expansion of Chinese TPS to target sectors
 - Discussions in Australia, Brazil, Canada, Taiwan, and elsewhere

Initial target sectors

	Trade intensity (%)	Global emissions (%)
Aluminum	20	3
Steel	18	8
Fertilizer	52	1
Cement	3	6
Electricity	2	33
Hydrogen	0.1	2



Data

Global data by plant for 2023

- Aluminum smelters from WoodMac (153 worldwide)
 - Public data and site visits
- Steel mills from Climate TRACE (892 worldwide)
 - Satellite and mill-level sensor data
- Production, capacity, costs, and emissions
 - Subnational carbon taxes, allowances, and exemptions

Production

Aluminum

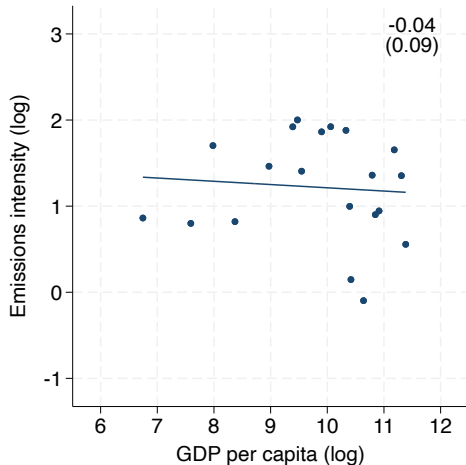
	Mt	%
China	49	58
India	5	6
Europe	5	5
USA	4	5
Russia	4	5
Rest of world	18	21

Steel

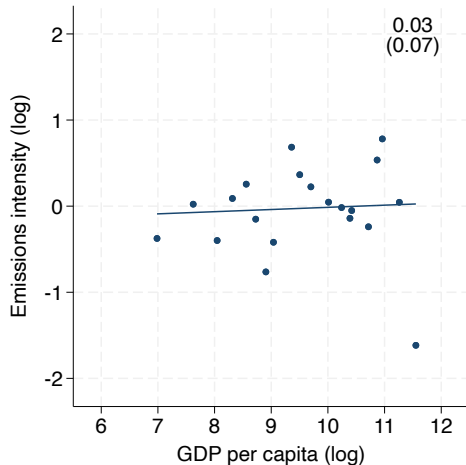
	Mt	%
China	860	51
Europe	153	9
Japan	88	5
USA	86	5
India	76	5
Rest of world	409	24

Emissions intensity is income-neutral

Aluminum



Steel



Emissions are not systematically higher in lower-income countries

- Even controlling for differences in production
 - Aluminum: electricity emissions are also income-neutral
 - Steel: basic chemical processes govern emissions intensity
- The pattern extends beyond aluminum and steel
 - Chemicals, petrochemicals, cement, and more

Emission sources

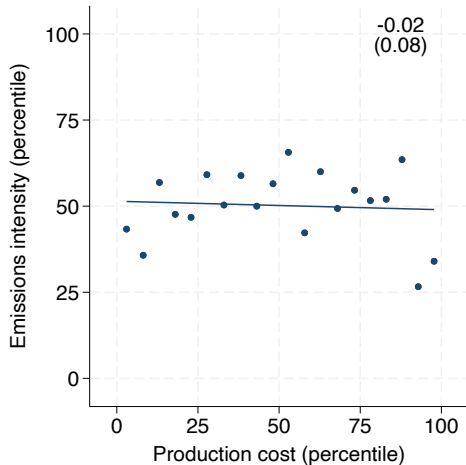
Controls

Electricity

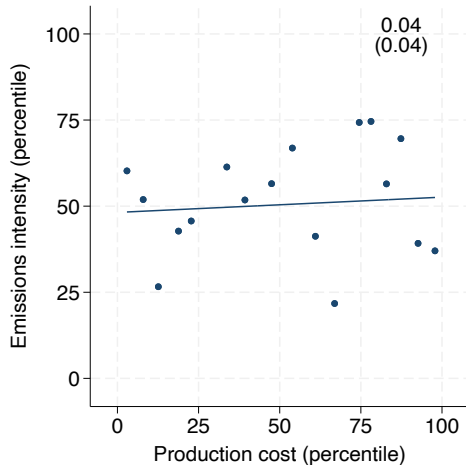
Other sectors

Emissions intensity is cost-neutral

Aluminum



Steel



Two facts shape CBAM incidence

- Emissions intensity is **income-neutral**
 - Carbon pricing does not systematically target lower-income countries
- Emissions intensity is **cost-neutral**
 - Carbon pricing creates new comparative advantage
- Much of the relevant variation is within countries
 - Plant-level measurement is central to policy design

Model

Environment

- Products $j \in \{\text{aluminum, steel}\}$ and countries k, ℓ
- Competitive plants choose three margins
 - ① How much to produce
 - ② Where to sell
 - ③ How much to abate
- Country markets clear at endogenous prices
 - Carbon policy affects production, trade, and emissions

Demand d_{jk}

$$\log d_{jk} = a_{jk} + b_j \log p_{jk}$$

- Country-product demand depends on local price p_{jk}
 - $b_j < 0$ is the product-specific demand elasticity
- Consumer surplus follows by integrating demand
 - Higher prices shift the burden toward consumers

Production s_{ijk}

$$u_{ijk}^l = v_{ijk} + \epsilon_{ijk}^l, \quad v_{ijk} = \beta_j(p_{ijk} - c_{ijk})$$

$$s_{ijk} = s_{ijk}^{\max} o_{ijk}, \quad o_{ijk} = \frac{\exp(v_{ijk})}{1 + \exp(v_{ijk})}$$

- Plants operate production lines with heterogeneous costs c_{ijk}
 - Capacity utilization o_{ijk} rises with profit margins
- Marginal costs rise as a plant approaches capacity
 - Marginal and average costs need not coincide

Trade shares σ_{ijk}^ℓ

$$\sigma_{ijk}^\ell = \frac{\exp(p_{ijk}^\ell / \theta_j)}{\sum_{\ell'} \exp(p_{ijk}^{\ell'} / \theta_j)}, \quad p_{ijk}^\ell = p_{j\ell} - \delta_{jk}^\ell - r_{ijk}^\ell$$

- Plants allocate sales across destination markets ℓ
 - Trade costs δ_{jk}^ℓ rationalize observed flows
- Regulation r_{ijk}^ℓ affects destination-specific net prices p_{ijk}^ℓ
 - Trade shares respond, subject to logit scale θ_j

Abatement of emissions e_{ijk}^ℓ

$$r_{ijk}^\ell = \min_e \{ \tau_{jk}^\ell e + f_{ijk}(e) \}, \quad e_{ijk}^\ell = \bar{e}_{ijk} \exp\left(\frac{\bar{\tau}_{jk} - \tau_{jk}^\ell}{\gamma_j}\right)$$

- Carbon prices τ_{jk}^ℓ encourage plants to reduce emissions intensity
 - Abatement becomes progressively more costly
- Observed emissions anchor each plant's abatement curve
 - γ_j governs the elasticity of emissions intensity

Carbon regulation

- Without border adjustment, tax depends on origin

$$\tau_{jk}^{\ell} = \tau_{jk}$$

- With border adjustment, destination ℓ fills the tax gap

$$\alpha_{jk}^{\ell} = \tau_{j\ell} - \tau_{jk}, \quad \tau_{jk}^{\ell} = \tau_{j\ell}$$

- Border revenue accrues to the importing government
 - Credits for origin taxes create a fiscal incentive to regulate

Border adjustment selects on emissions intensity

$$\frac{\partial \log \sigma_{ijk}^{\ell}}{\partial \tau_{j\ell}} = -\frac{(1 - \sigma_{ijk}^{\ell})e_{ijk}^{\ell}}{\theta_j} < 0$$

- High-emission plants shift sales away from CBAM markets
 - Supply falls and prices rise in the regulated market
- Foreign plants can instead abate to lower the border charge
 - Policy balances **abatement** against trade **reshuffling**

Estimation

Demand estimation

$$\log D_{jt} = a_j + a_t + b_j \log P_{jt} + \epsilon_{jt}$$

- World consumption and prices from 1976 to 2024
 - Product and year fixed effects
- Price instrument: Australia's share of global ore production
 - Large ore producer; less than 1% of global metal consumption

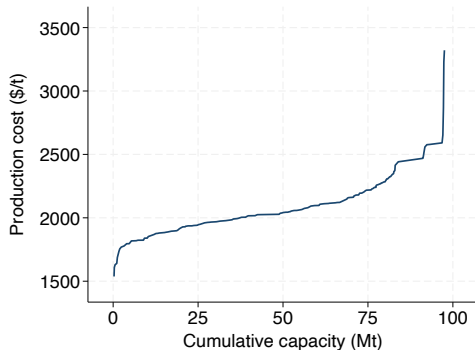
Demand is elastic

	Aluminum		Steel		Obs
	Estimate	SE	Estimate	SE	
1976 to 2024					
OLS	-0.40***	0.10	-0.61***	0.06	98
IV: Australian share of ore production	-1.69***	0.61	-1.34***	0.31	98
1998 to 2022					
OLS	-0.47***	0.12	-0.47***	0.08	50
IV: Australian share of ore production	-1.17***	0.41	-0.97***	0.27	50
IV: concentration of ore production	-0.86***	0.33	-0.81***	0.20	50
IV: both instruments	-0.97***	0.30	-0.80***	0.19	50

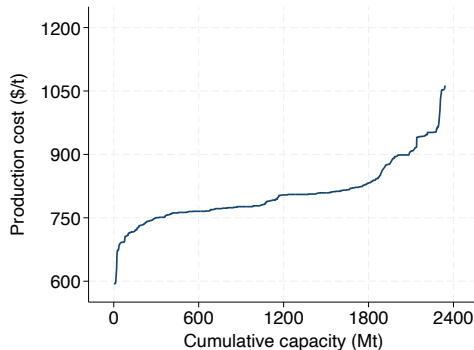
Elasticities: -1.69 for aluminum, -1.34 for steel

Average cost data

Aluminum



Steel



Cost curves $\neq$ supply curves

Supply estimation with cost data

$$MC_{ijk}^* - AC_{ijk} = -\frac{\log(1 - o_{ijk})}{\beta_j o_{ijk}}, \quad MC_{ijk}^* = p_{ijk}$$

- Average cost data capture marginal costs (Domínguez-lino et al. 2026)
 - Model structure maps the wedge using utilization o_{ijk}
- Cost and utilization identify β_j within product-country markets
 - Product-country fixed effects η_{jk} absorb prices and trade terms

Supply estimation with cost data

$$-\frac{\log(1 - o_{ijk})}{o_{ijk}} = \beta_j(P_j - \bar{\tau}_{jk}\bar{e}_{ijk} - AC_{ijk}) + \eta_{jk} + \xi_{ijk}$$

- Average cost data capture marginal costs (Domínguez-lino et al. 2026)
 - Model structure maps the wedge using utilization o_{ijk}
- Cost and utilization identify β_j within product-country markets
 - Product-country fixed effects η_{jk} absorb prices and trade terms

Supply is inelastic

	Aluminum			Steel		
	Estimate	SE	Obs	Estimate	SE	Obs
Elasticities						
With cost data	0.91**	0.36	153	0.01	0.16	881
No cost data	7.59	8.02	153	-0.41	1.04	881
Parameters						
With cost data	2.51**	0.99	153	0.02	0.51	881
No cost data	21.0	22.3	153	-1.30	3.33	881

Elasticities: 0.91 for aluminum, 0.01 for steel

Calibration

- Trade elasticity targets 8 (Caliendo and Parro 2015)
 - $\theta_{\text{aluminum}} = 282$ and $\theta_{\text{steel}} = 126$
- Abatement elasticity targets -0.3 (Sen and Vollebergh 2018)
 - $\gamma_j = 333$ for both metals
- Bilateral trade costs match observed destination shares

Counterfactuals

Policy simulations

- Europe raises its carbon tax to \$100 per ton of CO₂
 - From observed effective taxes with allowances and exemptions
- Compare carbon taxation with and without a CBAM
 - The difference isolates the effect of border adjustment
- Solve for market-clearing prices and trade flows under each policy
 - Compute emissions and welfare (CS , PS , G)

CBAM raises European prices, lowers foreign prices

(% change)	Without CBAM		With CBAM	
	Europe	Rest of world	Europe	Rest of world
Price	0.12	0.01	6.12	-0.50
Consumption	-0.20	-0.02	-7.98	0.70
Production	-0.47	0.01	-0.04	-0.05

- CBAM reallocates trade more than production (reshuffling)
 - Inelastic supply, elastic demand

CBAM boosts domestic competitiveness

Europe (\$1B)	Without CBAM	With CBAM	Difference
Consumer surplus	-0.20	-9.51	-9.31
Producer surplus	-13.5	-4.90	8.60
Government revenue	11.0	14.1	3.10
Welfare	-2.64	-0.29	2.35

- European producers gain, but European consumers lose (concentrated vs. diffuse)

CBAM curbs leakage, induces abatement

Emissions (Mt CO ₂)	Without CBAM	With CBAM	Difference
European consumption	-32.2	-91.8	-59.6
Foreign production	0.40	-11.5	-11.9
Global emissions	-37.3	-48.4	-11.1

- Global emissions reductions rise by 30%

CBAM can improve global social welfare

SCC	Global social welfare (\$1B)		
	Without CBAM	With CBAM	Difference
\$50	-0.69	-1.19	-0.50
\$100	1.18	1.23	0.05
\$200	4.91	6.07	1.16
\$500	16.10	20.59	4.49

- Net effect depends on abatement gains vs. reshuffling losses

CBAMs tilt incentives toward foreign regulation

- European CBAM penalizes high-emission exports to Europe
 - Non-regulation becomes less attractive
- Partner CBAM protects its own home market
 - Border revenue offsets domestic regulation costs
- A larger coalition reduces competitiveness concerns
 - Carbon taxes become strategic complements

Participation incentives vary across partners

	Welfare cost of regulation			
	Without CBAM (\$1B)	EU CBAM (%Δ)	Partner CBAM (%Δ)	With China (%Δ)
China	31.4	-1.12	-4.04	
India	3.20	-1.90	17.1	8.56
Japan	1.92	-18.0	-16.7	-28.9
Russia	1.42	-33.3	-31.7	-61.2
USA	1.25	-1.87	-80.3	-78.0
South Korea	1.21	-37.5	-16.8	-25.9

- CBAM effects reflect export orientation and home market size
- China effects illustrate gains from a larger coalition

Producer losses are consumer gains

Consumer surplus

Largest gains (\$1B)		Largest losses (\$1B)	
China	3.20	Germany	-1.97
India	1.56	Italy	-1.44
Japan	0.57	Spain	-0.80
Russia	0.54	Poland	-0.75
South Korea	0.48	France	-0.73

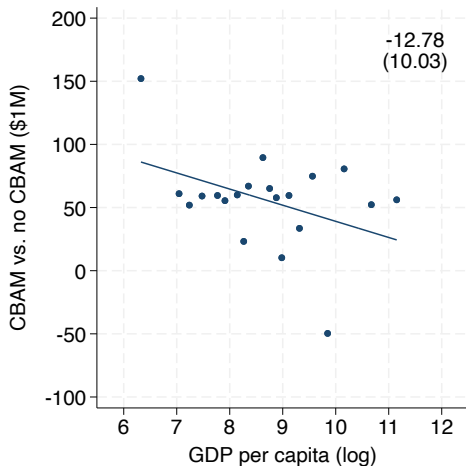
Producer surplus

Largest gains (\$1B)		Largest losses (\$1B)	
Germany	2.08	China	-3.80
Italy	1.45	India	-1.54
France	0.74	Ukraine	-1.39
Spain	0.61	Japan	-0.83
UK	0.40	Russia	-0.82

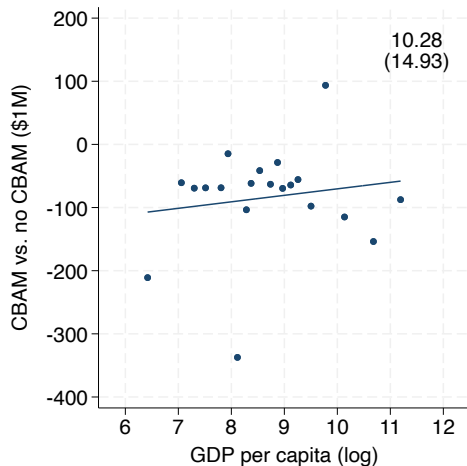
- High-emission producers lose, but foreign consumers gain (lower foreign prices)

Lower-income countries are not hit harder

Consumer surplus



Producer surplus



Coarse measurement weakens border adjustment

	Global emissions (Mt)	Global social welfare (\$1B)
Plant-specific measures	-48.4	1.23
Country-mode averages by metal	-40.9	0.73
Country averages by metal	-41.0	1.12
Global averages by metal	-41.8	1.08
No border adjustment	-37.3	1.18

- Group averages remove plant-level abatement incentives (reshuffling remains)

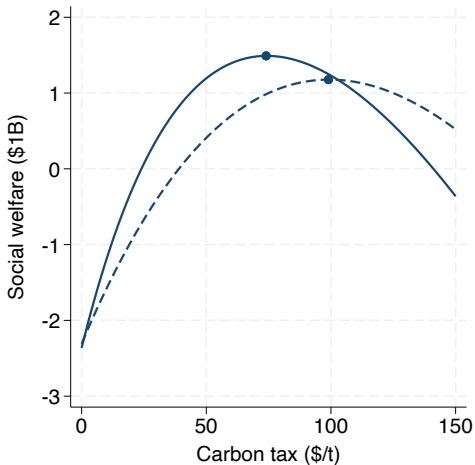
Partial adjustment can dominate full adjustment

	Global emissions (Mt)	Global social welfare (\$1B)
Full	-48.4	1.23
75%	-47.7	1.68
50%	-46.1	1.99
25%	-43.0	1.96
None	-37.3	1.18

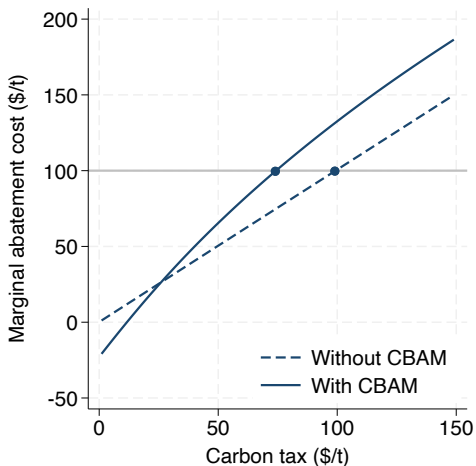
- Half adjustment raises social welfare to \$1.99B (preserve abatement, lessen reshuffling)

CBAM raises gains, reduces costs

Social welfare



Marginal abatement cost



When does a CBAM work best?

- Strong abatement responses favor border adjustment
 - Foreign plants clean up rather than reroute sales
- High trade openness favors reshuffling
 - Exporters can avoid the border charge more easily
- Plant-specific and partial adjustment improve the balance
 - Preserve green incentives while limiting deadweight loss

Conclusion

Summary

- CBAMs can make carbon regulation easier to sustain
 - Competitiveness, leakage, and participation improve
- Burden determined by emissions intensity, not income
 - Plant-level measurement and modeling matter

Appendix

Indirect emissions for aluminum, direct for steel

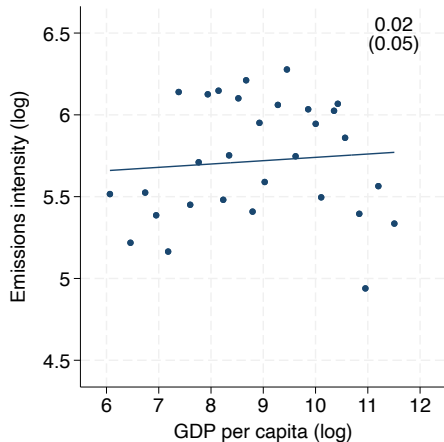
	Aluminum (%)	Steel (%)
Direct (Scope 1)	23.2	90.9
Indirect (Scope 2)	76.8	9.1

Controlling for production mode, scale, ownership, and age

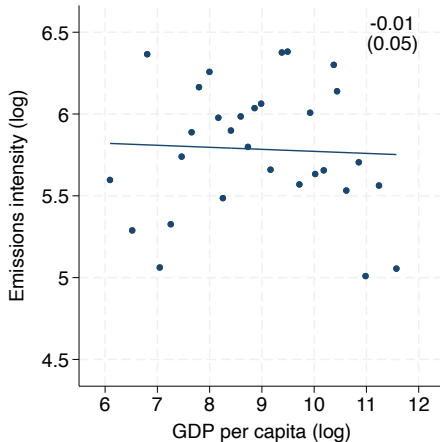
	Without controls			With controls		
	Estimate	SE	Obs	Estimate	SE	Obs
Aluminum						
Total emissions	−0.04	0.10	38	−0.07	0.11	34
Direct (Scope 1)	−0.08**	0.03	38	−0.02	0.01	34
Indirect (Scope 2)	−0.01	0.18	38	−0.09	0.17	34
Steel						
Total emissions	0.07	0.05	77	−0.04	0.04	77
Direct (Scope 1)	0.14*	0.07	77	0.00	0.06	77
Indirect (Scope 2)	−0.24***	0.05	77	−0.09*	0.05	77

Electricity emissions are income-neutral

Electricity Maps



Ember



Income neutrality beyond aluminum and steel

